(Please write your Enrollment Number) Enrollment No. 03201062021

MID-TERM EXAMINATION

(CBCS)

B. Tech (ECE/CSE), 2nd Semester

(June, 2022)

Subject Code: BAS 104	Subject: Applied Physics –II

Time : 1 ½ Hours Maximum Marks : 30

Note:Q. 1 is compulsory. Attempt any two questions from the rest.

Q1		(2.5*4)
	(a) Draw the I-V curve of a p-n junction diode. Clearly indicate the various regions.	
	(b) How does the conductivity of a metal vary with temperature? How is different for a semiconductor?	
	(c) Define Fermi Energy level.	
	(d) What is volt equivalent of Temperature?	
Q2		(5,5)
	(a) Derive the expression $E_F = E_C - kTln\left(\frac{N_c}{N_D}\right)$	
	(b) Explain different current densities in a semiconductor.	
Q3		(5,5)
	(a) Write the four Maxwell equations in differential form.	
	(b) Derive the expression for skin depth.	
Q4		(5,5)
	(a) What are the boundary conditions for electric and magnetic fields?	
	(b) Explain Poynting theorem qualitatively.	